

2025 EDITION

FREE

MCP

THE ILLUSTRATED GUIDEBOOK



Daily Dose of
Data Science

Avi Chawla & Akshay Pachaar
DailyDoseofDS.com

How to make the most out of this book and your time?

The reading time of this book is about 3 hours. But not all chapters will be of relevance to you. This 2-minute assessment will test your current expertise and recommend chapters that will be most useful to you.

Are you MCP-aware?

Answer 8 yes/no questions and we'll email you the list of chapters that must read to improve your MCP skillset.

Take the Assessment Now!



Start The Assessment

Start the Assessment

Scan the QR code below or open this link to start the assessment. It will only take 2 minutes to complete.



<https://bit.ly/mcp-assessment>

Table of contents

Section #1) Model Context Protocol.....	3
1.1) What is MCP?.....	4-5
Introduction.....	4-5
1.2) Why was MCP created?.....	6-8
The problem.....	6-7
The solution.....	7-8
1.3) MCP Architecture Overview.....	9-11
Host.....	9
Client.....	10
Server.....	11
1.4) Tools, Resources and Prompts.....	12-18
Tools.....	12
Resources.....	14
Prompts.....	15
Section #2) MCP Projects.....	19
2.1) 100% local MCP client.....	20
2.2) MCP-powered Agentic RAG.....	25
2.3) MCP-powered Financial Analyst.....	29
2.4) MCP-powered Voice Agent.....	34
2.5) A unified MCP server.....	39
2.6) MCP-powered shared memory for Claude Desktop and Cursor.....	43
2.7) MCP-powered RAG over complex docs.....	47
2.8) MCP-powered Synthetic Data Generator.....	51
2.9) MCP-powered Deep Researcher.....	57
2.10) MCP RAG over videos.....	63
2.11) MCP-powered Audio Analysis Toolkit.....	69